Standard Operating Procedure for Resolving ASIC Memory Parity Errors and Line Card Crashes on Arista Access Switches (BAS-ACC-06)

Node: BAS-ACC-06 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_2783

Description

The Arista Access Switch (BAS-ACC-06) deployed at the Basavanagudi POP exhibits a hardware/environmental fault characterized by ASIC memory parity errors and line card crashes, triggering Alarm Code ALARM_2783 with a Minor severity. This failure domain is rooted in the underlying architecture of the Arista Access Switch, where the ASIC memory parity errors are a manifestation of bit-flip errors in the SRAM (Static Random Access Memory) or DRAM (Dynamic Random Access Memory) used for packet forwarding and buffering. Furthermore, line card crashes can occur due to overheating, power supply redundancy loss, or software/hardware incompatibilities. In the context of this particular role, the Arista Access Switch is responsible for providing Layer 2 and Layer 3 access services, and any hardware or environmental faults can disrupt the forwarding plane, impacting service delivery and network availability. To mitigate this risk, it is essential to promptly diagnose and resolve such faults through diligent troubleshooting and repair procedures.

Pre-requisite Checks

- Check: Verify the power supply redundancy status using the 'show power' command to ensure that the switch has sufficient power to operate without any issues.

- Check: Verify the temperature and humidity levels in the Basavanagudi POP using the environmental monitoring system to ensure that the conditions are within the recommended operating range for the Arista Access Switch.

- Check: Verify the redundant link status using the 'show interface' command to ensure that the link is stable and not experiencing any issues.

Potential Root Causes

Cause	Probability
ASIC Memory Parity Errors	High
Line Card Crash	Medium
Power Supply Redundancy Loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show tech-support detail, Expected Output: Detailed system information, including ASIC memory parity error logs and line card crash logs., Description: This command provides a comprehensive view
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of the system logs, allowing for the identification of potential
issues related to ASIC memory parity errors and line card crashes.

Command: show system resource, Expected Output: System resource
utilization, including memory usage and CPU utilization., Description:
This command provides insight into the current system resource
utilization, helping to determine if the issue is related to resource
constraints.

Command: show interface detail, Expected Output: Interface status,
including link speed and duplex mode., Description: This command
provides information about the interface status, helping to determine
if the issue is related to link connectivity or configuration.
```

Resolution Steps

Step 1: Step: Clear ASIC Memory Parity Error Logs, Command: clear system log, Description: This command clears the system log, removing any existing records of ASIC memory parity errors, allowing for a fresh start and potentially resolving the issue.

Step 2: Step: Replace Faulty Line Card, Command: shutdown interface , Description: This command shuts down the faulty line card, preventing further packet forwarding and buffering issues. Once the faulty line card is replaced, the interface can be brought back up.

Step 3: Step: Reboot the Switch, Command: reboot, Description: This command reboots the switch, allowing for a fresh start and potentially resolving the issue.

Post-Check Verification

```
Command: show system resource, Expected Output: System resource
utilization, including memory usage and CPU utilization., Description:
This command provides insight into the current system resource
utilization, helping to determine if the issue has been resolved.

Command: show interface detail, Expected Output: Interface status,
including link speed and duplex mode., Description: This command
provides information about the interface status, helping to determine
if the issue has been resolved.
```

Roll-back Procedures

- Step: Restore Original Configuration, Command: load merge , Description: This command restores the original configuration from a backup file, reversing any changes made during the troubleshooting and repair process.

- Step: Reboot the Switch, Command: reboot, Description: This command reboots the switch, allowing for a fresh start and ensuring that the system returns to its original state.